
Ukulele Robot With Mobile Application

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Index Terms: Ukulele, Robot, Mobile Application.

1. Mid-term Progress

The ukulele robot is equipped with four set of arms in order to control the four corresponding string. Currently we assembled two set of arms on a wooden board and planned the overall architecture of the robot.

1.1. Hardware

Key to this project is the ability to play the ukulele which is allowed by the arms being equipped with functional elements attachments and incorporating the slider mechanism into the project to provide better movement to the arm.

To ensure consistency in performance, the ukulele, sliders, and motor drivers will be fastened on an wooden board. This enhances effective mechanical interaction and ensures that there will not be disturbances while performing. Due to the defined placement of these constituents, the body natural movements are acted to ensure that the sound remains undisturbed after featuring machine interactions.

1.1.1. Electronic component

The reciprocating 42 step slider we are using are of belt transmission, this physical feature of the sliders that allows for linear movement is important in ensuring that any unnecessary motion of the slider due to vibrations can not happen. Preventing digression of the slider ensures that a constant operation is maintained.

On top of each set of slider a NEMA 17 stepper motor are placed with a geared rack which provides movement along the z axis, ensures that the robot can form proper chord shapes on the strings.

We connect the slider and stepper motor with its corresponding driver, hence both wired to a UNO board for in-set communication. Then all four UNO will be connected to Raspberry Pi for further communication.

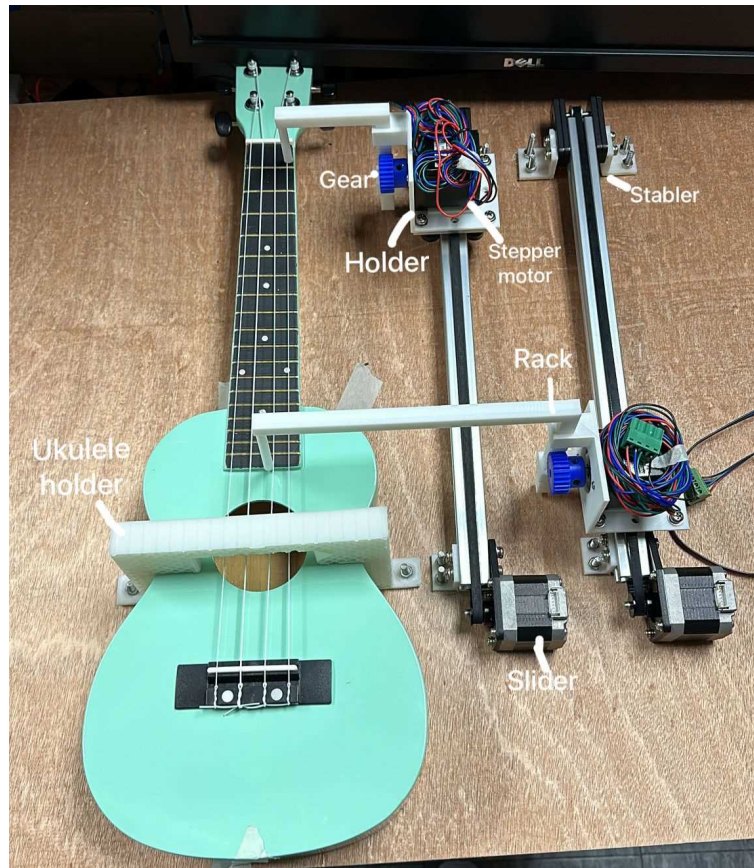


Fig. 1. Top view of the ukulele robot with each component labeled.

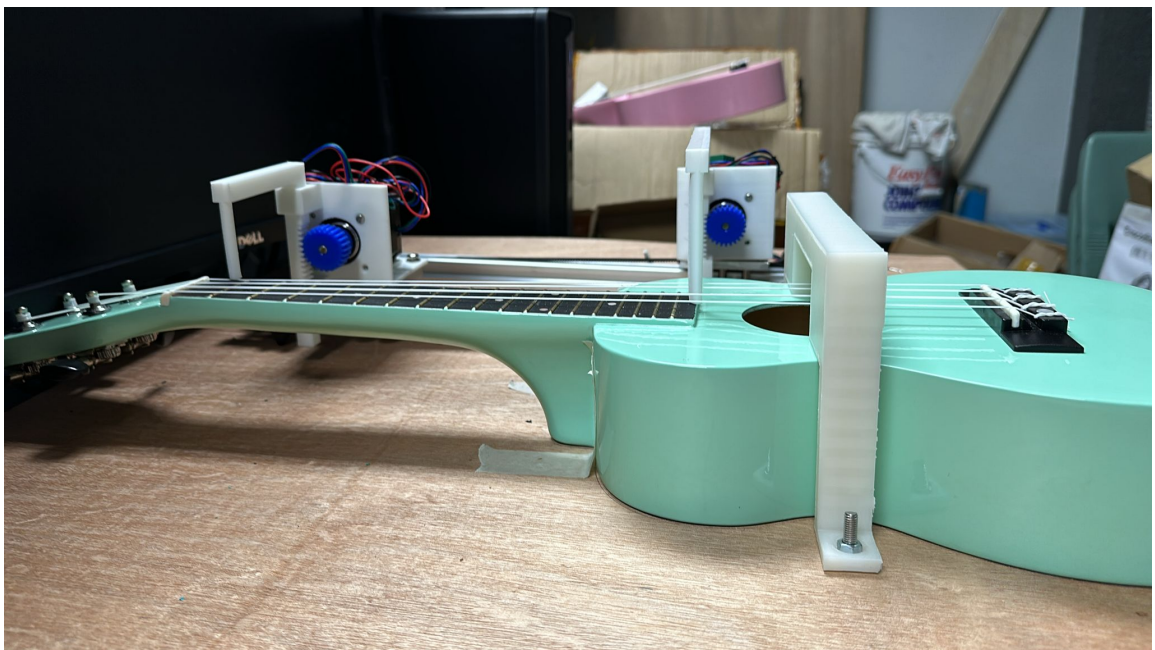
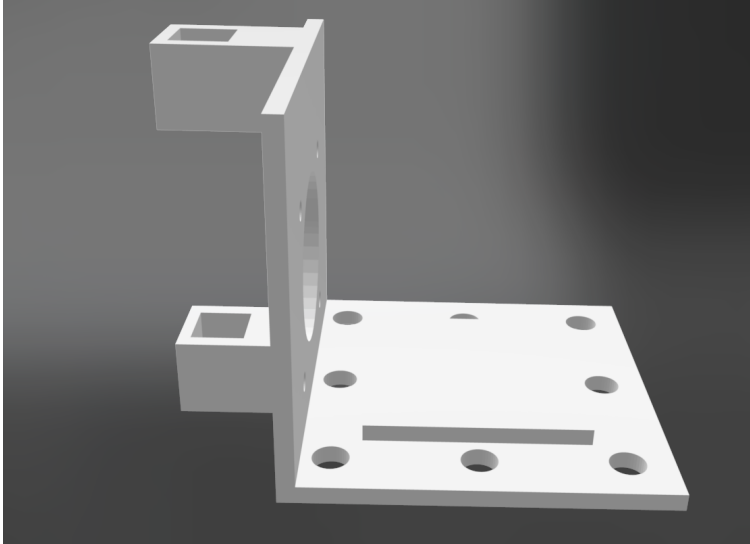


Fig. 2. Side view of the ukulele robot showcasing the structure of the gear and rack.

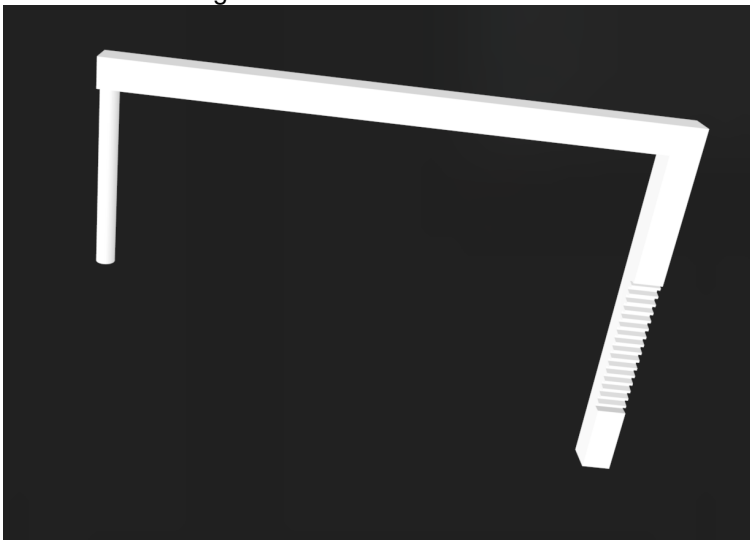
1.1.2. 3D print

3D printing is widely used in this project to tailor the component that specify the size and design we need.

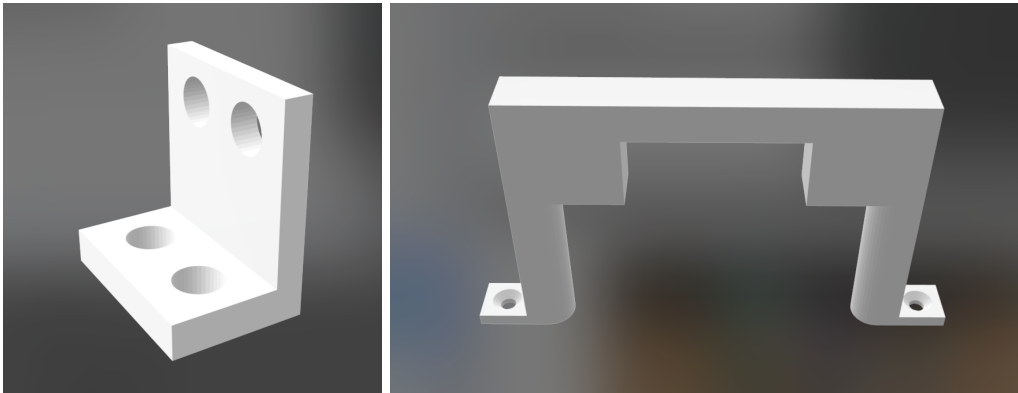
The holder is designed to place on top of the slider to hold the motor driver, gear and rack in place. One of the key design of the holder is to leave room for rack to move while ensuring the space is not too large that the rack will wiggle during vertical movement. With the design as following:



The rack, on the other hand, is designed to control all excessive movements allowing only manipulations in z axis, and thus provides a basis for string manipulation. The design draft is shown as following:



Stabler and ukulele holder are printed to hold the slider and the ukulele respectively to the board to avoid unwanted deviation. The designs are shown below:



1.2. Software

1.2.1. Introduction

The software part contains three parts, including the Arduino Uno, Raspberry Pi, and the mobile application.

For the Arduino Uno part, we use it to control the motors. We connect each Uno board using I2C connection by assigning each Uno board a unique network address. In total, we will use 4 Uno boards for controlling the vertical and horizontal move of the rack, also an additional board for plucking the strings.

For the Raspberry Pi part, we use it to receive Bluetooth signal from the mobile device and send signals to the Arduino Uno boards. It acts as a bridge between the user end and the hardware part.

For the mobile application part, we plan to use a framework that is universal for multi operating systems. React Native is currently picked as it has a large community and it can develop Android and iOS applications at the same time.

1.2.2. Code

For the Arduino Uno part, we will calibrate each motor first. For example, we will move the motor to the end to ensure it has the same position every time we start up. We will use an infrared sensor to detect its position. After that it will turn into discovery mode and let the others to send commands to it. We have normalized the input format to make instructions more readable.

For the Raspberry Pi part, it will first detect all devices reachable in the network connected by the I2C port. Then it will compare it to a dictionary that contains the device information. After confirming all devices are workable, it will start a pairing mode in Bluetooth. It will automatically accept all incoming pairing and the user can choose to disconnect other devices. We allow multiple device connections because we would wish to always keep the device in pairing mode, so we have no need to remove the pairing every time if we want another device to connect it. When the user is connected, they can send commands to the Arduino Uno through the Raspberry Pi. The details of commands are not implemented yet.

For the mobile application part, most features are not implemented yet. However, it should be able to send commands and check the session with Raspberry Pi.

2. Next step

2.1. Hardware

After focusing on the string pressing mechanism, our next goal is to design for the sweeping mechanism which we plan to use a servomotor with a blade to mimic human chord sweeping. The design of the ukulele holder will be modified to integrate the servomotor set into the robot.

There will be final adjustment on the design of those 3D printing to ensure the spacing among different component are accurate, all components can hold in position, avoid any digression. Furthermore, after assuring the functionality of the robot is satisfied, we would like to rearrange layout of the robot and modify the appearance of each 3D component to make it more appealing.

2.2. Software

We will start the development of mobile application part. Our first target is let the guitar robot to play chords. To achieve this part, we have to complete the development to Bluetooth pairing on both Raspberry Pi and mobile devices. After that, we have to calculate the moving distance of each slider and rack for each chord, then we can control the exact distance on Raspberry Pi. After that, when our mobile device send the specific chord that the user wants to play, the Raspberry Pi can directly control the hardware to play the note.

Afterwards, we would plan to implement the score feature that the user can input the entire song to it. We can extend the score feature and let it play consecutive notes to form a score. However, this part maybe implement in term two.